



# Point Neuron Embedded Kalman Filter (PNEKF)

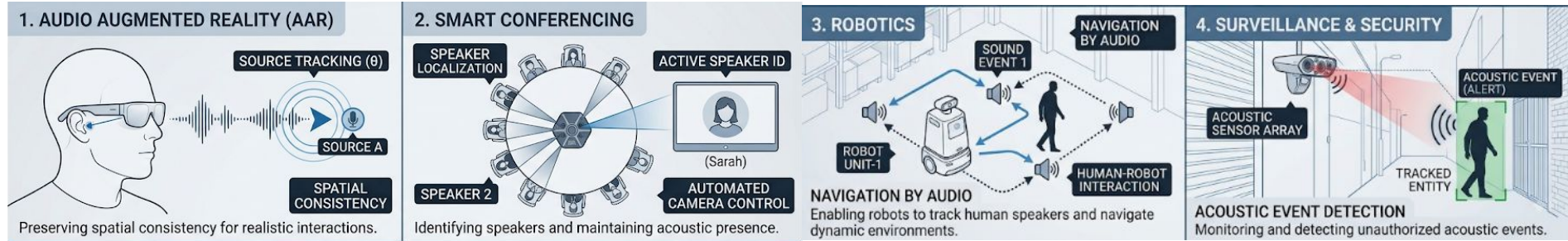
A Narrowband Sound Source Tracking Algorithm in Reverberant Rooms

**Vishwanath Vinod\*** , Shaoheng Xu<sup>+</sup>, Prasanga Samarasinghe<sup>+</sup>, Amy Bastine<sup>+</sup>,  
Thushara D Abhayapala<sup>+</sup>

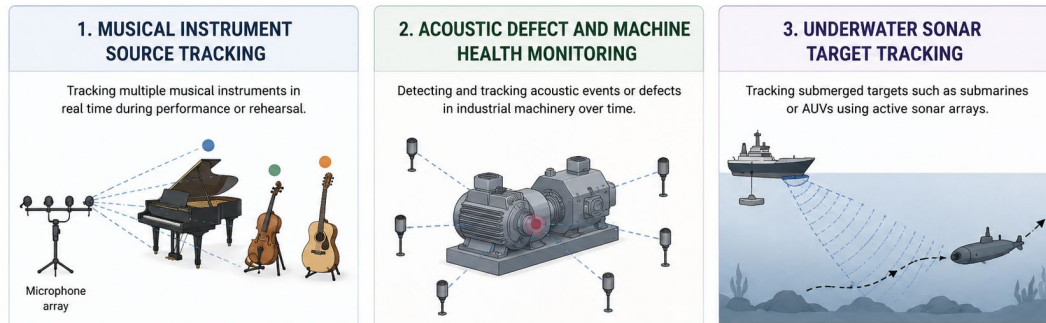
\* Indian Institute of Technology Madras, Chennai, India.

<sup>+</sup> Australian National University, Canberra, Australia

# Sound Source Tracking (SST)



Although broadband sources are common in many practical scenarios, narrowband SST remains highly relevant in applications involving machine noises, musical instruments and wireless communications.



# Limitations of Existing Methods

1. Most methods smooth grid-based sound source localization (SSL) estimates at every time frame with filters to obtain a continuous trajectory.

This can lead to grid-mismatch errors when the source lies between grid points, limiting accuracy.

*Ex:*

**MUSIC + KF/SMC-PHD:** Multiple Signal Classification-based localization with Kalman filtering or Sequential Monte Carlo Probability-Hypothesis Density filtering

**TDOA + PF/ADPF/MMKF:** Time Difference of Arrival estimates from Generalized cross correlation with Particle Filters, Adaptive Distributed Particle filters or Multi-Model Kalman filters

**SRP + PF:** Steered -response-power beamforming with Particle filters

2. 3-D tracking of narrowband sound sources in reverberant environments is not well studied as it is hindered by limited frequency content and reflected image sources.

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| MUSIC + KF/SMC-PHD                                                                                                                        | TDOA + PF/ADPF/MMKF                                                                                                                                                         | SRP +PF                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Multiple Signal Classification localization<br>+<br>Kalman filtering / Sequential Monte Carlo<br>Probability-Hypothesis Density filtering | Time Difference of Arrival estimates from<br>Generalized cross correlation<br>+<br>Particle Filters, Adaptive Distributed Particle<br>filters or Multi-Model Kalman filters | Steered -response-power<br>beamforming<br>+<br>Particle filters |

2. 3-D tracking of narrowband sound sources in reverberant environments is not well studied (**limited frequency content and reflected image sources**).

# Point Neurons

Sound fields must satisfy the free space Green's function, a fundamental solution of the Helmholtz wave equation.

$$\mathcal{G}(\mathbf{x} \mid \mathbf{y}, k) = \frac{e^{ik\|\mathbf{x}-\mathbf{y}\|_2}}{4\pi\|\mathbf{x}-\mathbf{y}\|_2}$$

where  
 $\mathbf{x} \in \mathbb{R}^3$  : observation point,  
 $\mathbf{y} \in \mathbb{R}^3$  : source location,  
 $k \in \mathbb{R}$  : wavenumber.

$k = \frac{2\pi f}{c}$  where  $f$  is frequency of the sound source.

An additional normalization is also included to unify the treatment of both near-field and far-field sources.

$$\text{PN}(\mathbf{x} \mid \mathbf{y}, k) = \|\mathbf{y}\|_2 e^{-ik\|\mathbf{y}\|_2} \frac{e^{ik\|\mathbf{x}-\mathbf{y}\|_2}}{4\pi\|\mathbf{x}-\mathbf{y}\|_2}$$

While PINNs enforce this physical constraint through an additional PDE loss term, Point Neuron Learning aims to embed it directly into the network architecture.

Specifically, each fundamental unit (**point neuron**) acts as a **virtual acoustic source** and outputs its effective **sound pressure** at the input location using two **learnable parameters**:

1. **Weight:** Intensity/Strength of the virtual source,
2. **Bias:** Location of the virtual source.

# How we Propose to Tackle it?

We introduce the Point Neuron Embedded Kalman Filter (PNEKF), a novel Kalman filter variant that integrates the point neuron framework for 3D tracking of multiple narrow band sound sources.

1. **Grid-mismatch errors:** PNEKF uses SSL only once for initialization and thereafter operates directly on microphone measurements, enabling continuous 3D tracking.
2. **Advantages of point neurons:**
  - a. complete differentiability, allowing efficient backpropagation for optimization
  - b. clear physical interpretation of weights and biases as acoustic parameters.
3. **Robust tracking in reverberant settings:** By embedding point neurons along room boundaries, PNEKF dynamically simulates image sources induced by reverberation.
4. **Empirical Validation:** Improved accuracy over baseline methods for single- and multiple-SST tracking across a range of frequencies and trajectories.

# Problem Setting: Setup

The sound pressure field at the  $\mathbf{Q}$  microphones is reconstructed using  $\mathbf{J}$  point neurons: the first  $\mathbf{N}$  are placed at the estimated source positions to capture the direct component, while the remaining  $\mathbf{J}-\mathbf{N}$  are uniformly distributed along the room boundaries to model reverberation.

$$\begin{aligned}
 P_q^t(f) &= \sum_{j=1}^J w_j^t \cdot \mathbf{PN}(\mathbf{B}_j^t, \mathbf{m}_q, f) \\
 &= \sum_{j=1}^J w_j^t \cdot \|\mathbf{B}_j^t\|_2 e^{-ik\|\mathbf{B}_j^t\|_2} \frac{e^{ik\|\mathbf{m}_q - \mathbf{B}_j^t\|_2}}{4\pi\|\mathbf{m}_q - \mathbf{B}_j^t\|_2},
 \end{aligned}$$

$w_j^t \in \mathbb{R}$ : weight/strength of the  $j^{\text{th}}$  point neuron,  
 $\mathbf{B}_j^t \in \mathbb{R}^3$ : bias/position of the  $j^{\text{th}}$  point neuron,  
 $\mathbf{s}_n^t$ : true position of the  $n^{\text{th}}$  source,  
 $\mathbf{m}_q$ : position of the  $q^{\text{th}}$  microphone,  
 $P_q^t(f)$ : sound pressure at microphone  $q$  at frequency  $f$ .

$\mathbf{PN}(\mathbf{B}_j^t, \mathbf{m}_q, f)$ : nonlinear response of point neuron  $j$  at microphone  $q$ ,

$\mathbf{D}^t \in \mathbb{C}^{Q \times J}$ : forward model matrix stacking all microphone-point neuron responses.

# Problem Setting: System Model & Objective

$$\begin{aligned}
 \text{State space: } \mathbf{X}^t &= [\mathbf{B}_1^t \quad \cdots \quad \mathbf{B}_N^t \quad \mathbf{v}_1^t \quad \cdots \quad \mathbf{v}_N^t]^\top, & \text{source positions and velocities,} \\
 \text{Measurement space: } \mathbf{Z}^t &= [P_1^t(f) \quad \cdots \quad P_Q^t(f)]^\top, & \text{microphone sound-pressure recordings.}
 \end{aligned}$$

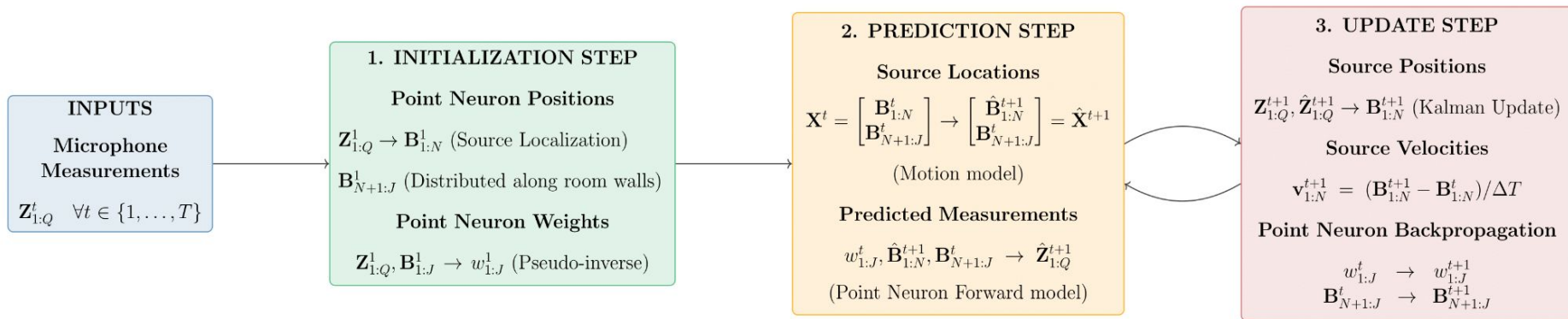
$$\begin{aligned}
 \text{Process noise from source accelerations: } & \mathbf{y}^t \sim \mathcal{N}(\mathbf{0}, \mathcal{Y}), \\
 \text{Measurement noise from microphone uncertainty: } & \mathbf{r}^t \sim \mathcal{N}(\mathbf{0}, \mathcal{R}).
 \end{aligned}$$

The estimated positions of the  $N$  sources at each time frame are modelled by the positions of their corresponding point neurons.

$$\hat{\mathbf{s}}_n^t = \mathbf{B}_n^t, \quad \forall n \in \{1, 2, \dots, N\}$$



# Algorithm Overview



PNEKF initializes point neuron weights and positions at  $t=1$  using SSL. It then alternates between prediction and update steps, operating directly on microphone measurements.

In prediction, source locations are propagated via a linear motion model and predicted measurements are computed.

In update, Kalman update estimates source positions, while velocities are computed from successive source positions, and backpropagation updates weights and positions.

# PNEKF: Initialization

The tracking process begins with the initialization of the weights and positions of the point neurons at  $\mathbf{t}=\mathbf{1}$ .

**Positions:** The sound sources are localized using an SSL algorithm, and the  $\mathbf{N}$  point neurons corresponding to them are placed at these estimated positions. The remaining  $\mathbf{J}-\mathbf{N}$  point neurons are uniformly distributed along planes aligned with the room boundaries.

**Weights:** Given the microphone measurements and point neuron positions, the weights can be initialized using the Moore–Penrose pseudoinverse of the  $\mathbf{D}^1$  matrix.

$$[w_1^1, \dots, w_J^1]^\top = (\mathbf{D}^1)^\dagger \mathbf{Z}^1,$$

# PNEKF: Prediction

The prediction step of PNEKF employs a linear motion model using the Kalman filter prediction equations.

$$\begin{aligned} \text{Predicted Next State: } \hat{\mathbf{X}}^{t+1} &= \mathbf{A}\mathbf{X}^t + \mathbf{C}y^t, \\ \text{State Error Covariance: } \hat{\mathbf{F}}^{t+1} &= \mathbf{A}\mathbf{F}^t\mathbf{A}^\top + \mathcal{Y}. \end{aligned}$$

We define the state transition and noise matrices as:

$$\mathbf{A} = \begin{bmatrix} \mathbf{I}_N & \mathbf{O} & \Delta T \mathbf{I}_N \\ \mathbf{O} & \mathbf{I}_{J-N} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{I}_N \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \frac{1}{2}(\Delta T)^2 \mathbf{I}_N \\ \mathbf{O} \\ \Delta T \mathbf{I}_N \end{bmatrix}.$$

$\mathbf{A}$  : evolves source positions and velocities, while keeping reverberant states fixed,

$\mathbf{C}$  : injects acceleration noise to account for deviations from constant-velocity motion.

**Motion Dynamics:** The reverberant point neuron locations are modelled as static, while source point neurons evolve with velocity  $\mathbf{v}_{1:N}^t$ , with both their positions and velocities perturbed by acceleration.

# PNEKF: Prediction

Conventional KF computes predicted measurements by linearizing the relation between measurement and state space using the Jacobian  $\mathbf{H}^t$ .

$$\hat{\mathbf{Z}}^{t+1} = \mathbf{H}^t \hat{\mathbf{X}}^{t+1}, \quad \mathbf{H}^t = \begin{bmatrix} \frac{\partial P_1^t(f)}{\partial \mathbf{B}_1^t} & \cdots & \frac{\partial P_1^t(f)}{\partial \mathbf{B}_J^t} & \frac{\partial P_1^t(f)}{\partial \mathbf{v}_1^t} & \cdots & \frac{\partial P_1^t(f)}{\partial \mathbf{v}_N^t} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial P_Q^t(f)}{\partial \mathbf{B}_1^t} & \cdots & \frac{\partial P_Q^t(f)}{\partial \mathbf{B}_J^t} & \frac{\partial P_Q^t(f)}{\partial \mathbf{v}_1^t} & \cdots & \frac{\partial P_Q^t(f)}{\partial \mathbf{v}_N^t} \end{bmatrix}.$$

However, PNEKF employs the full non-linear forward model of Point Neuron Learning, to predict the sound pressure using the predicted point neuron positions.

$$\hat{\mathbf{Z}}_q^{t+1} = \sum_{j=1}^J w_j^t \cdot \text{PN}(\hat{\mathbf{B}}_j^{t+1}, \mathbf{m}_q, f).$$

Since the measurements are complex-valued, we use their real-valued augmented forms, denoted by :

$$\mathbf{M} = [\Re(\mathbf{M}) \ \Im(\mathbf{M})], \quad \mathbf{M} \in \{\mathbf{Z}^t, \hat{\mathbf{Z}}^t, \mathbf{H}^t\}.$$

# PNEKF: Update

We compute the Kalman gain to balance predicted and observed measurements.

$$\mathbf{K}^t = \hat{\mathbf{F}}^t (\mathbf{H}^t)^\top \left( \mathbf{H}^t \hat{\mathbf{F}}^t (\mathbf{H}^t)^\top + \mathcal{R} \right)^{-1}.$$

**Positions:** The estimated positions of the  $\mathbf{N}$  moving sound sources are obtained using the Kalman update rule to the  $\mathbf{N}$  source point neurons.

$$\mathbf{B}_{1:N}^{t+1} = \hat{\mathbf{B}}_{1:N}^{t+1} + \mathbf{K}_{[1:3N,:]}^t (\mathbf{Z}^{t+1} - \hat{\mathbf{Z}}^{t+1}).$$

**Velocities:** Since measured sound pressure depends only on source positions, velocities are not directly observable and are instead computed implicitly from successive position estimates

$$\mathbf{v}_{1:N}^{t+1} = \frac{(\mathbf{B}_{1:N}^{t+1} - \mathbf{B}_{1:N}^t)}{\Delta T}.$$

# PNEKF: Update

**Backpropagation:** The positions of the **J**-**N** reverberation point neurons and the weights of all the **J** point neurons are updated using the backpropagation step of Point Neuron Learning.

The backpropagation loss function is defined as:

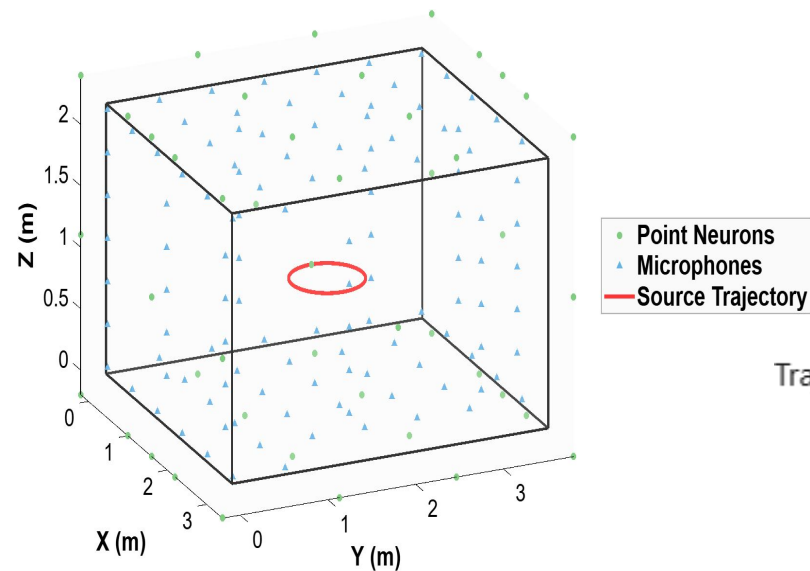
$$\mathcal{L}(w_{1:J}^t, \mathbf{B}_{N+1:J}^t) = \lambda \sum_{j=1}^J \|w_j^t\|_1 + \sum_{q=1}^Q \|\mathcal{Z}_q^{t+1}(f) - P_q^{t+1}(f)\|_2^2.$$

The first term enforces weight sparsity, and the second minimizes sound field reconstruction error at microphones.

Here, the expected microphone measurement is computed using the updated source locations, the current reverberant point neuron locations and point neuron weights:

$$\mathcal{Z}_q^{t+1}(f) = \sum_{j=N+1}^J w_j^t \mathbf{PN}(\mathbf{B}_j^t, \mathbf{m}_q, f) + \sum_{j=1}^N w_j^t \mathbf{PN}(\mathbf{B}_j^{t+1}, \mathbf{m}_q, f). \quad (1)$$

# Empirical Results: Setting



3.6 m  $\times$  3.2 m  $\times$  2.2 m room

Region of Interest (ROI): 1 m  $\times$  1 m  $\times$  1 m at the centre.

The moving sound sources are constrained to this ROI.

The Image-Source Method is used to compute the acoustic transfer function between each source-microphone pair at every time frame  $t$  and all simulations are conducted in the frequency domain.

Trajectories are discretized into  $T = 500$  time frames separated by  $\Delta T = 1s$ .

Reverberation coefficients: [0.7, 0.8, 0.65, 0.7, 0.5, 0.6]

Maximum reflection order: 10      Microphones with SNR: 20dB

Diagonal error covariance matrices initialization:

$\mathbf{F}^1$  :  $10^{-6}$  for positions,  $10^{-5}$  for velocities,

$\mathcal{R}$  :  $10^{-2}$  for measurement noise,

$\mathcal{Y}$  :  $10^{-10}$  for process noise.

# Empirical Results: Design

## Initialization of Sound Source Positions:

$J - N = 78$  point neurons are placed on planes parallel to the room boundaries, with a **0.2 m** outward offset (prevent coinciding with microphones) to model reverberation.

A modified **Orthogonal Matching Pursuit** (OMP) algorithm is used on a grid of candidate positions inside the ROI to iteratively estimate the **N** source positions from the de-reverberated measurements.

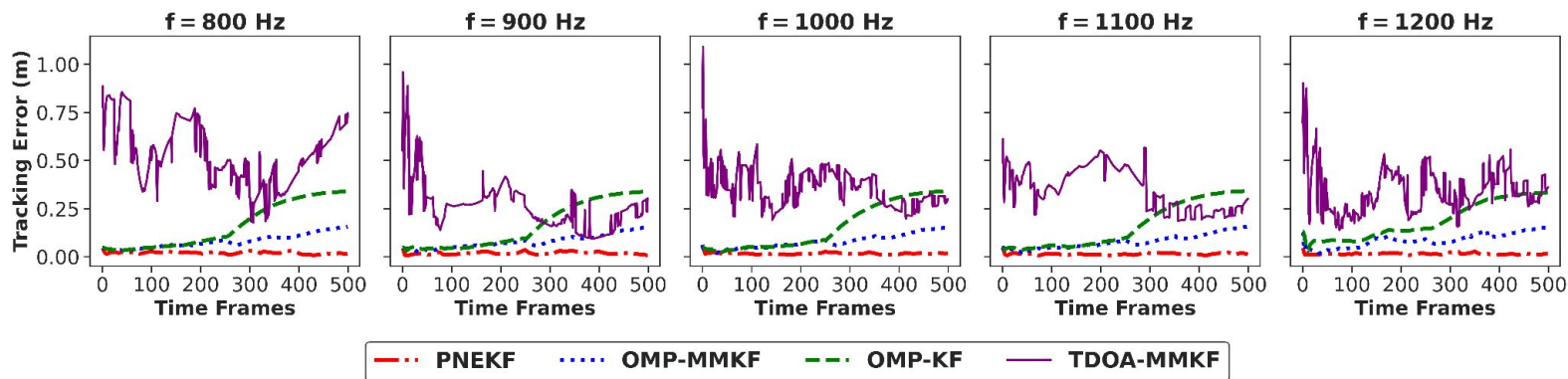
## Evaluation Metric:

At each time instant  $t$ , the tracking error for any source  $n \in \{1, \dots, N\}$  is defined as the RMSE between the estimated and ground-truth sources averaged over all  $M$  experiments.

$$\mathbf{TE}_n(t) = \sqrt{\frac{1}{M} \sum_{m=1}^M \|\hat{\mathbf{s}}_n^{t,(m)} - \mathbf{s}_n^{t,(m)}\|_2^2}.$$



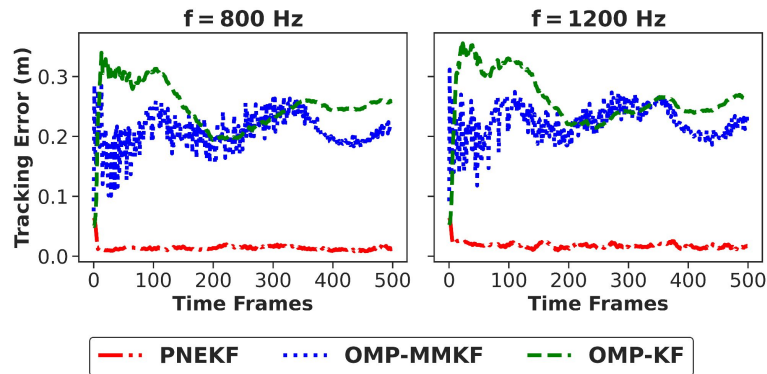
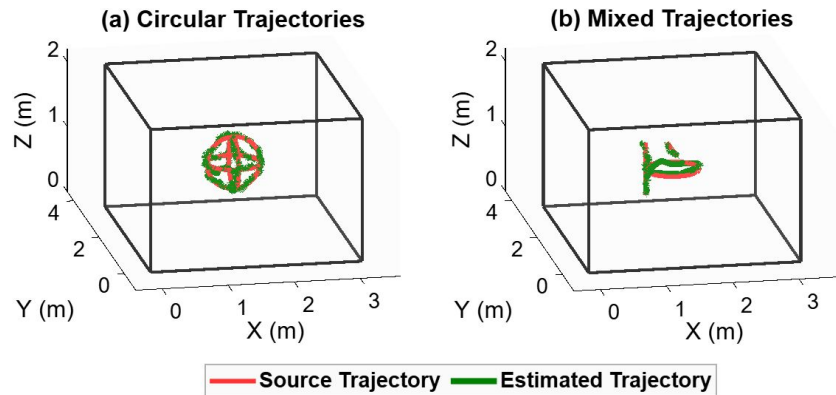
# Empirical Results: Single SST



Comparison of the proposed PNEKF with TDOA-MMKF, OMP-MMKF, and OMP-KF in the single-SST scenario, evaluated with  $Q=95$  microphones.

**PNEKF consistently outperforms all baselines across the tested source frequencies, maintaining near-zero tracking error.**

# Empirical Results: Multiple SST



Multiple-SST results using PNEKF with  $Q=119$  microphones.

Left: Qualitative visualization of PNEKF tracking performance at  $f=1200$  Hz for 3 sources following circular and mixed trajectories.

Right: Quantitative comparison with OMP-MMKF and OMP-KF at  $f=800$  Hz and  $f=1200$  Hz using average tracking error across all sources and both trajectories.

**PNEKF can accurately track multiple source trajectories within the ROI, demonstrating its scalability.**

# Conclusion

The proposed PNEKF framework for tracking narrowband sources in reverberant rooms:

- **Operates directly on microphone measurements**, enabling continuous 3D tracking without grid mismatch.
- **Achieves significantly lower tracking errors** across frequencies and trajectories than existing baselines.
- **Scales effectively to multiple sources**, enabling accurate simultaneous trajectory tracking

## Future Work

- Extending to broadband sources using broadband Point Neuron Learning to reduce microphone count.
- Generalization to an unknown number of sources.
- Eliminating SSL-based initialization for fully measurement-driven tracking.
- Real-world experimental validation.

# Contacts:



Vishwanath Vinod,  
Undergraduate (B.Tech),  
Department of Electrical Engineering @ IIT Madras.  
Mail: [vishwanathvinod2004@gmail.com](mailto:vishwanathvinod2004@gmail.com)

Thushara D. Abhayapala,  
Professor, Audio and Signal Processing Group  
Mail: [thushara.abhayapala@anu.edu.au](mailto:thushara.abhayapala@anu.edu.au)

